

MODULE - IV

Relativistic Mechanics

The classical mechanics can be applied if velocity of a particle is low ($v \ll c$), but it fails at $v \approx c$.
Theory of Relativity can be applied for all velocities from 0 to c .

Frame of Reference

A system of coordinate axes ^{with} ~~which~~ respect to which the position of a particle or event is described is called frame of reference. It must be rigid.

Example-1: Cartesian Coordinate System, in which three coordinates x, y & z are used to describe the position.

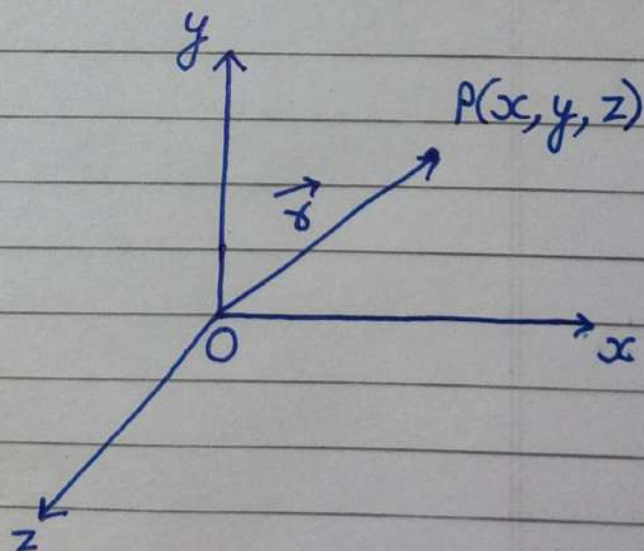
Position Vector,
$$\vec{r} = \hat{i}x + \hat{j}y + \hat{k}z$$

Velocity, $\vec{v} = \frac{d\vec{r}}{dt}$

$$\vec{v} = \hat{i} \frac{dx}{dt} + \hat{j} \frac{dy}{dt} + \hat{k} \frac{dz}{dt}$$

Acceleration, $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$

$$\vec{a} = \hat{i} \frac{d^2x}{dt^2} + \hat{j} \frac{d^2y}{dt^2} + \hat{k} \frac{d^2z}{dt^2}$$



Example-2: Space Time Frame is a coordinate system with four coordinates x, y, z & time ' t ' is called space-time frame.

Inertial Frame:

In this frame, Newton's laws are valid. This frame is at rest or moves with constant velocity.

Non-Inertial Frame:

In this frame, Newton's laws are not valid. They are either rotating or accelerating.

Q. Is Earth an Inertial Frame?

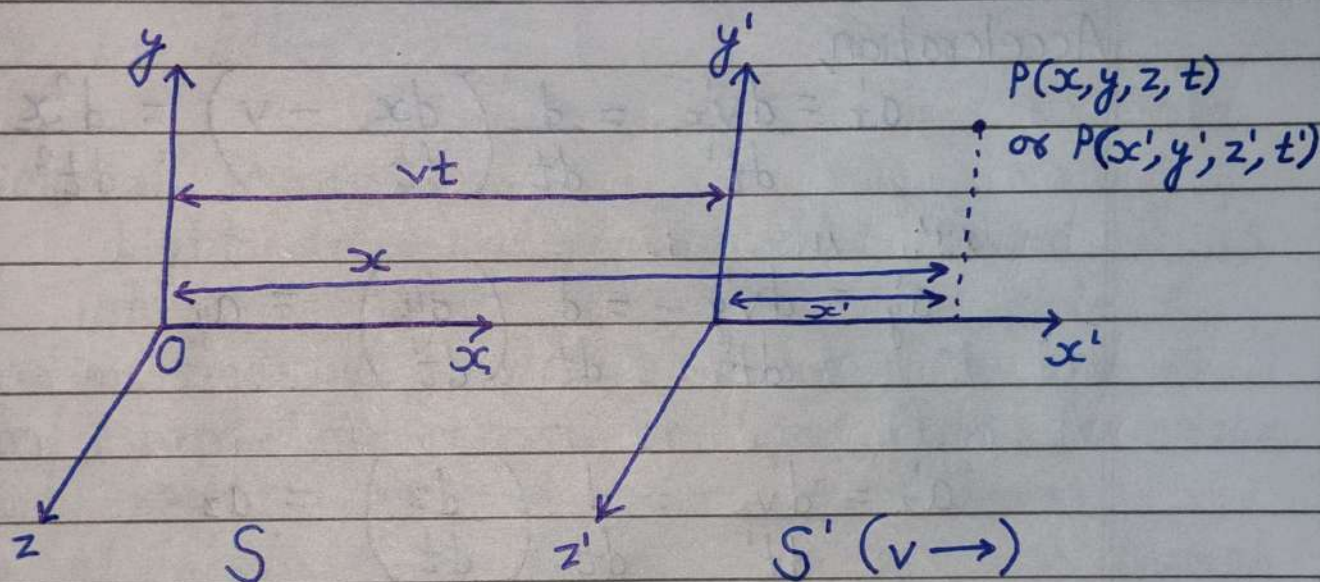
⇒ Earth is non-inertial frame because it rotates.

Since, the speed is very low therefore it can be considered as inertial frame of reference.

Galilean Transformation Equations

Consider two inertial frames S & S' . The equations relating two set of coordinates are called Transformation Equation.

At $t = t' = 0$, both frames are coincident. As time passes, S is at rest and S' moves with constant velocity v in the x -direction.



From diagram,

$$x = x' + vt$$

$$\Rightarrow x' = x - vt \quad \text{--- (1)}$$

$$y' = y \quad \text{--- (2)}$$

$$z' = z \quad \text{--- (3)}$$

$$t' = t \quad \text{--- (4)}$$

Velocity,

$$v_x' = \frac{dx'}{dt'} = \frac{d(x - vt)}{dt} = \frac{dx}{dt} - v$$

$$v_y' = \frac{dy'}{dt'} = \frac{dy}{dt}$$

$$v_z' = \frac{dz'}{dt'} = \frac{dz}{dt}$$

Acceleration,

$$a_x' = \frac{dv_x'}{dt'} = \frac{d}{dt} \left(\frac{dx}{dt} - v \right) = \frac{d^2x}{dt^2} - 0 = a_x$$

$$a_y' = \frac{dv_y'}{dt'} = \frac{d}{dt} \left(\frac{dy}{dt} \right) = a_y$$

$$a_z' = \frac{dv_z'}{dt'} = \frac{d}{dt} \left(\frac{dz}{dt} \right) = a_z$$

$$\Rightarrow \vec{a}' = \vec{a}$$

i.e. Acceleration ~~is~~ is invariant under Galilean Transformation.

Michelson - Morley Experiment

Concept of Ether:

A hypothetical medium Ether was assumed to be present everywhere in space. Ether must be rigid, massless, invisible and transparent.

Imp. Objective:

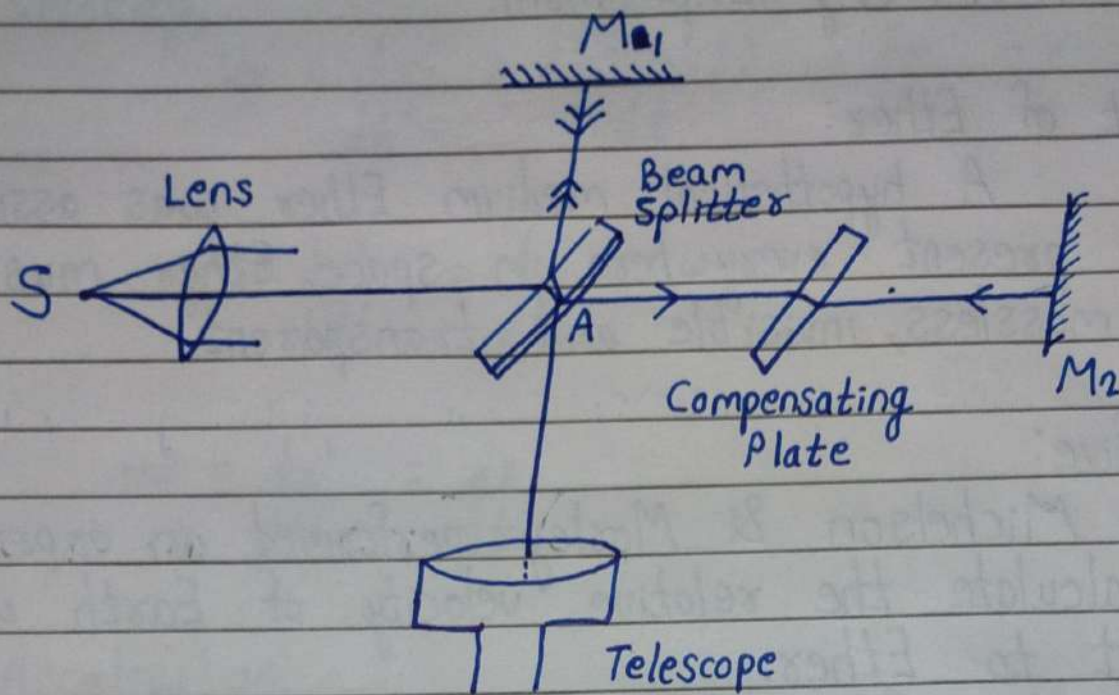
Michelson & Morley performed an experiment to calculate the relative velocity of Earth with respect to Ether.

Experimental Arrangement:

Light from mono-chromatic source S is divided into two parts by beam-splitter plate A. The reflected light travels to mirror M_1 and is reflected back. The transmitted light goes to mirror M_2 and reflects back. The two light beams overlap at point A so fringes can be seen through telescope.

The reflected light travels beam-splitter ^{thrice} ~~twice~~ while the transmitted light only ones. To compensate this path difference, a compensating plate is used which is identical to beam-splitter plate.

The whole assembly was mounted on a stone-slab floating in mercury.



Theory:

From figure,

$$AM_1 = AM_2 = D$$

Let, T_1 is the time taken by light to travel from A to M_1 and T_2 to travel from M_1 to A .

$$T_1 = \frac{D}{c-v} \quad T_2 = \frac{D}{c+v}$$

Total time, $T = T_1 + T_2$

$$\Rightarrow T = \frac{D}{c-v} + \frac{D}{c+v} = \frac{2Dc}{c^2 - v^2}$$

Optical path, $x_1 = Tc$

$$\Rightarrow x_1 = \frac{2Dc^2}{c^2 - v^2}$$

$$\Rightarrow x_1 = \frac{2Dc^2}{c^2(1 - v^2/c^2)}$$

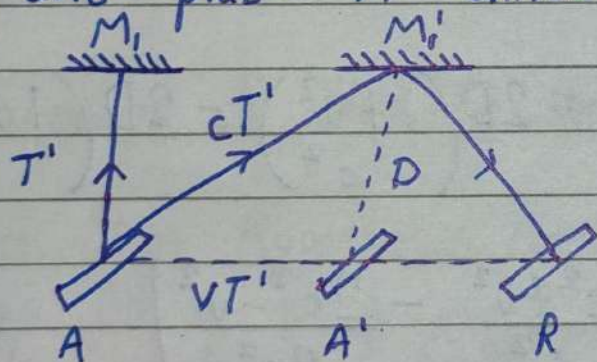
$$\Rightarrow x_1 = 2D \left(1 - \frac{v^2}{c^2}\right)^{-1}$$

Using Binomial Theorem and Neglecting higher orders

$$\left[(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \dots \right]$$

$$x_1 = 2D \left(1 + \frac{v^2}{c^2}\right) \text{ ————— (1)}$$

Let, T' be the time taken by light to travel from A to M_1 . During this time, mirror moves to position M_1' and plate A shifts to position A' .



$$c^2 T'^2 = v^2 T'^2 + D^2$$

$$\Rightarrow T'^2 (c^2 - v^2) = D^2$$

$$\Rightarrow T'^2 = \frac{D^2}{c^2 - v^2}$$

$$\Rightarrow T' = \frac{D}{\sqrt{c^2 - v^2}}$$

$$\text{Total time} = 2T'$$

$$\text{Optical path, } x_2 = 2T'c$$

$$\Rightarrow x_2 = \frac{2Dc}{\sqrt{c^2 - v^2}}$$

$$\Rightarrow x_2 = \frac{2Dc}{c\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow x_2 = 2D \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$$

Applying Binomial theorem and neglecting higher order terms,

$$x_2 = \left(1 + \frac{v^2}{2c^2}\right) \cdot 2D \quad \text{--- (2)}$$

$$\text{Path diff.} = x_1 - x_2 = 2D \left(1 + \frac{v^2}{c^2}\right) - 2D \left(1 + \frac{v^2}{2c^2}\right)$$

$$\Rightarrow \text{Path diff.} = 2D \left[\frac{c^2 + v^2}{c^2} - \frac{2c^2 - v^2}{2c^2} \right]$$

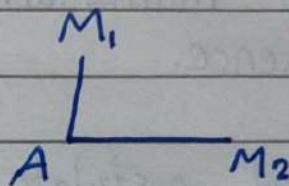
$$\Rightarrow \text{Path diff.} = \frac{2D}{2c^2} [2c^2 + 2v^2 - 2c^2 - v^2]$$

$$\Rightarrow \text{Path diff.} = \frac{Dv^2}{c^2}$$

Now, the apparatus is rotated through 90° , so that optical path AM_1 is longer than AM_2 by the same amount. Therefore, total path diff. = $\frac{2Dv^2}{c^2}$.

If this path diff. corresponds to shifting of n -fringes, then $\frac{2Dv^2}{c^2} = n\lambda$.

For $\lambda = 5000 \text{ \AA}$, $D = 10 \text{ m}$
 $\Rightarrow n = 0.4$ fringes.



Negative Result:

Michelson & Morley expected a fringe shift of 0.4 fringes. But no fringe-shift was observed in any direction. The negative result of this experiment strike on the roots of Ether Hypothesis.

Explanation of Negative Result:

- (i) Ether Drag Hypothesis: According to Michelson, the earth drags Ether along with it. There is no relative motion between Earth & Ether. Therefore, no fringe shift is observed.
- (ii) Lorentz-Fitz Gerald Contraction Hypothesis: According to Lorentz, the length of a moving body is contracted by a factor $\sqrt{1 - \frac{v^2}{c^2}}$ in the direction of motion.

Therefore, path diff. is zero and no fringe-shift is observed.

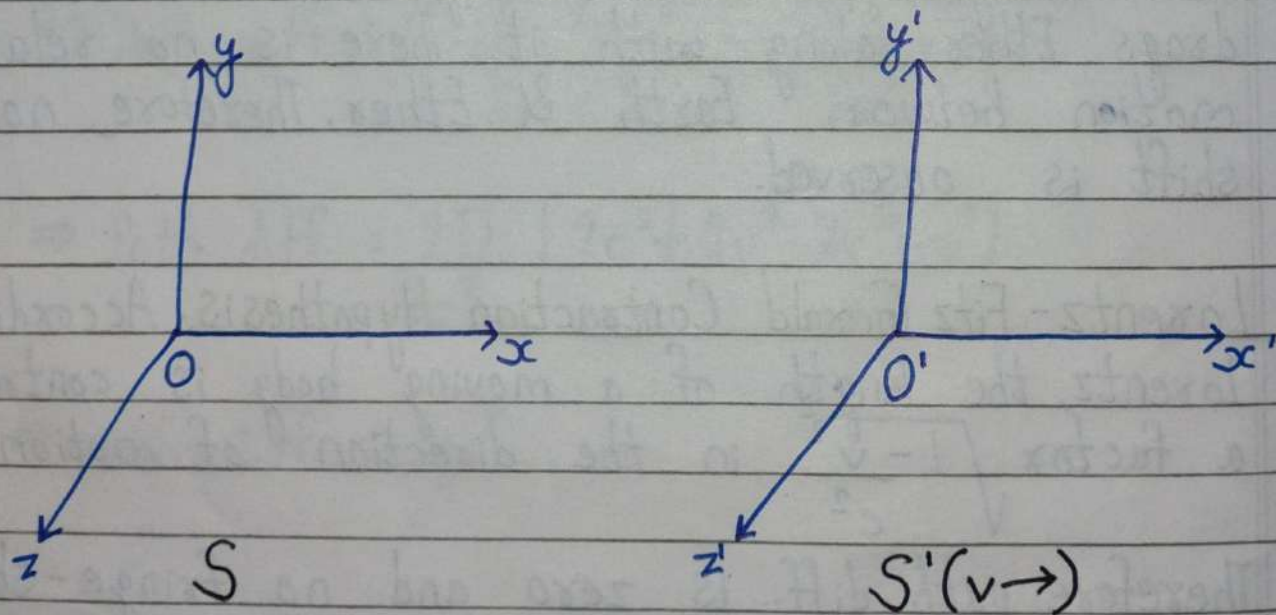
Imp (iii) Constancy of Speed of Light: According to Einstein, speed of light is constant and does not depend upon motion of source, medium or observer.

Einstein's Postulates

1. First postulate states that, the laws of physics have same mathematical form in every inertial frame of reference.
2. Second postulate states that, speed of light is constant and does not depend upon motion of source, medium or observer.

Lorentz Transformation Equations

Consider two inertial frames S & S' . At $t=t'=0$, both frames were coincident. Frame S is at rest and S' moves with constant velocity v in positive x -direction.



$$x' \propto (x - vt)$$

$$\Rightarrow x' = \gamma (x - vt) \text{ ————— (1)}$$

where,

$\gamma \rightarrow$ Constant of Proportionality.

Using First Postulate,

$$x = \gamma (x' + vt')$$

$$\text{————— (2)}$$

Putting x' in eqⁿ. (2),

$$x = \gamma [\gamma (x - vt) + vt']$$

$$\Rightarrow \frac{x}{\gamma} = \gamma x - \gamma vt + vt'$$

$$\Rightarrow vt' = \frac{x}{\gamma} - \gamma x + \gamma vt$$

$$\Rightarrow t' = \frac{x}{\gamma v} - \frac{\gamma x}{v} + \gamma t$$

$$\Rightarrow t' = \gamma t - \frac{\gamma x}{v} + \frac{x}{\gamma v}$$

$$\Rightarrow t' = \gamma t - \frac{\gamma x}{v} \left(1 - \frac{1}{\gamma^2} \right) \text{ ————— (3)}$$

Using First Postulate,

$$t = \gamma t' + \frac{\gamma x'}{v} \left(1 - \frac{1}{\gamma^2} \right) \text{ ————— (4)}$$

Let, a flash bulb emits radiation at $t = t' = 0$ from the common origin. Using second postulate, the distance travelled by light is given by,

$$x = ct$$
$$\& x' = ct'$$

Putting in eq^{ns} (1) & (2),

$$ct' = \gamma(ct - vt) = \gamma t (c - v) \text{ ————— (5)}$$

$$ct = \gamma(ct' + vt') = \gamma t' (c + v) \text{ ————— (6)}$$

Multiply eq^{ns}. (5) & (6),

$$c^2 \cancel{t} t' = \gamma^2 \cancel{t} t' (c^2 - v^2)$$

$$\Rightarrow \gamma^2 = \frac{c^2}{c^2 - v^2} = \frac{c^2}{c^2 (1 - v^2/c^2)}$$

$$\Rightarrow \gamma^2 = \frac{1}{1 - v^2/c^2}$$

$$\Rightarrow \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \text{ ————— (7)}$$

Putting value of γ in eqⁿ. (1), we get,

$$\boxed{x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}} \text{ ————— (8)}$$

$$\therefore \gamma^2 = \frac{1}{1 - v^2/c^2}$$

$$\Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{\gamma^2}$$

$$\Rightarrow 1 - \frac{1}{\gamma^2} = \frac{v^2}{c^2}$$

Putting values of γ & $\left(1 - \frac{1}{\gamma^2}\right)$ in eqⁿ. (3),

$$t' = \frac{t - \frac{x}{v} \times \frac{v^2}{c^2}}{\sqrt{1 - v^2/c^2}}$$

$$\boxed{t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}}} \quad \text{--- (9)}$$

Also,

$$y' = y \quad \text{--- (10)}$$

$$z' = z \quad \text{--- (11)}$$

Equations 8, 9, 10 & 11 are Lorentz Transformation Equations.

Inverse Lorentz Equations

We have,

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

Putting value of γ in eq^{ns}. (2) & (4), we get,

$$\cancel{t} \quad x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}$$

$$t = \frac{t' + vx'/c^2}{\sqrt{1 - v^2/c^2}}$$

$$y = y'$$

$$z = z'.$$

Q. Show that at low velocity Lorentz Transformation reduces to Galilean Transformation?

We have,

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}} \quad \& \quad t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}$$

if $v \ll c$, $v^2/c^2 \rightarrow 0$ & $v/c^2 \rightarrow 0$, we get,

$$x' = x - vt$$

$$y' = y$$

$$z' = z$$

$$t' = t.$$

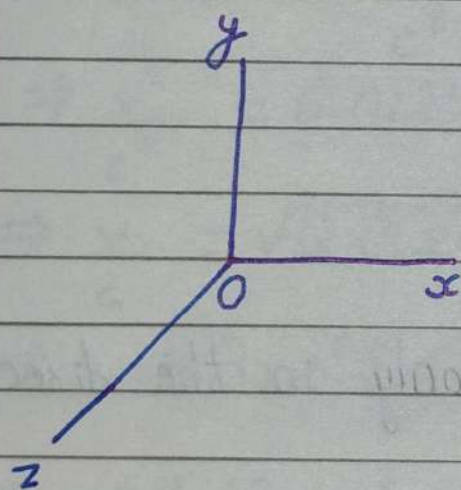
V. Imp. # Length Contraction

Consider a rod AB at rest in frame S' which is moving with constant velocity v in positive x -direction. The proper length of rod can be written as:-

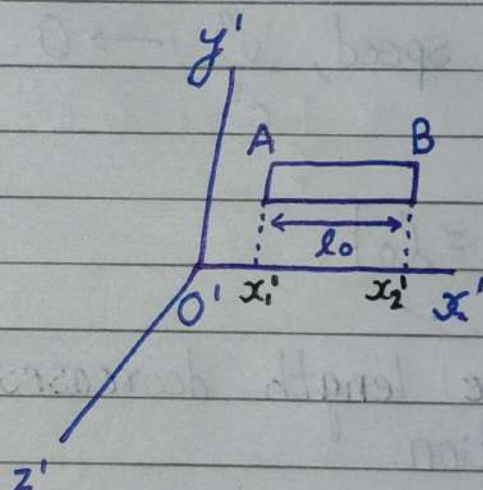
$$\text{Proper Length, } l_0 = x_2' - x_1' \quad \text{--- (1)}$$

At the same time, the length of rod w.r. to frame S which is at rest is given by:-

$$\text{Moving Length, } l = x_2 - x_1 \quad \text{--- (2)}$$



S



$S'(v \rightarrow)$

Lorentz Eq^{ns},

$$x_1' = \frac{x_1 - vt}{\sqrt{1 - v^2/c^2}}$$

$$x_2' = \frac{x_2 - vt}{\sqrt{1 - v^2/c^2}}$$

Putting values of x_1' & x_2' in eqⁿ. (1),

$$l_0 = \frac{x_2 - vt}{\sqrt{1 - v^2/c^2}} - \frac{x_1 - vt}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow l_0 = \frac{x_2 - x_1}{\sqrt{1 - v^2/c^2}}$$

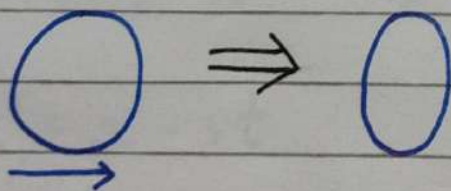
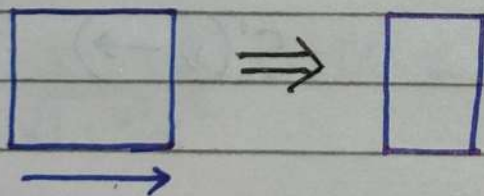
$$\Rightarrow l_0 = \frac{l}{\sqrt{1 - v^2/c^2}} \quad [\text{From eqⁿ. (2)}]$$

$$\Rightarrow \boxed{l = l_0 \sqrt{1 - \frac{v^2}{c^2}}} \quad [l < l_0]$$

At low speed, $\frac{v^2}{c^2} \rightarrow 0$

$$\boxed{l = l_0}$$

Note: The length decreases only in the direction of motion.



Q. How fast would a rocket have to go relative to an observer for its length to be contracted to 99% of its own length?

Solⁿ:

$$l = 99\% \text{ of } l_0 = 0.99 l_0$$
$$v = ?$$

We have,

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$\Rightarrow 0.99 l_0 = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$\Rightarrow 0.9801 = 1 - \frac{v^2}{c^2}$$

$$\Rightarrow \frac{v^2}{c^2} = 0.0199$$

$$\Rightarrow \frac{v}{c} = \sqrt{0.0199}$$

$$\Rightarrow v = 3 \times 10^8 \times \sqrt{0.0199}$$

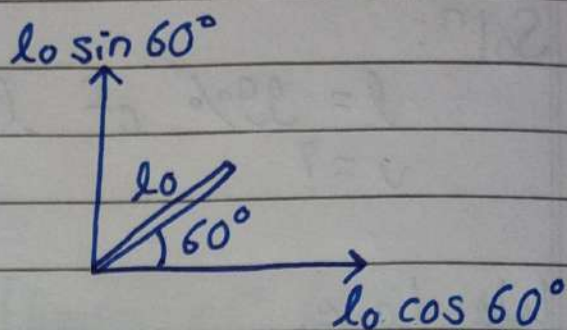
$$\Rightarrow v = 4.23 \times 10^7 \text{ m/s.}$$

Q. Calculate the percentage contraction of a rod moving with velocity $0.8c$ in a direction inclined at 60° to its own length?

Solⁿ:

$$v = 0.8c$$

$$\theta = 60^\circ$$



We have,

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Now,

$$l_x = l_0 \cos 60 \sqrt{1 - \left(\frac{0.8c}{c}\right)^2}$$

$$= l_0 \times \frac{1}{2} \sqrt{1 - 0.64}$$

$$= \frac{l_0}{2} \sqrt{0.36}$$

$$= \frac{0.6 l_0}{2}$$

$$\Rightarrow l_x = 0.3 l_0$$

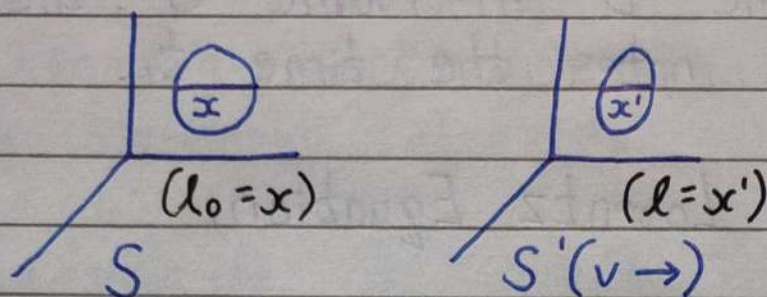
$$\text{Also, } l_y = l_0 \sin 60 = 0.866 l_0$$

$$\therefore l = \sqrt{l_x^2 + l_y^2} = 0.916 l_0$$

$$\% \text{ contraction} = \frac{l_0 - l}{l_0} \times 100 = 8.4\%$$

Q. Show that the circle $x^2 + y^2 = a^2$ in frame S appears to be an ellipse in frame S' which is moving with velocity v relative to S ?

Solⁿ:



We have,

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$
$$\Rightarrow x' = x \sqrt{1 - \frac{v^2}{c^2}}$$

$$\Rightarrow x = \frac{x'}{\sqrt{1 - v^2/c^2}}$$

$$\& y = y'$$

$$\therefore x^2 + y^2 = a^2$$

$$\Rightarrow \frac{x'^2}{1 - v^2/c^2} + y'^2 = a^2$$

$$\Rightarrow \frac{x'^2}{a^2(1 - v^2/c^2)} + \frac{y'^2}{a^2} = 1$$

$$\Rightarrow \frac{x'^2}{b^2} + \frac{y'^2}{a^2} = 1 \quad \left[a^2 \left(1 - \frac{v^2}{c^2} \right) = b^2 (\text{say}) \right]$$

This is the equation for Ellipse.

Time Dilation

Let us consider a clock at point x' in frame S' . Clock is at rest in frame S' . When an observer finds the time t_1' in frame S' , the observer in frame S notes the time t_1 .

Using Inverse Lorentz Equation,

$$t_1 = \frac{t_1' + vx'/c^2}{\sqrt{1 - v^2/c^2}}$$

After some time, the observer in frame S' notes time t_2' while the observer in frame S notes time t_2 .

$$t_2 = \frac{t_2' + vx'/c^2}{\sqrt{1 - v^2/c^2}}$$

In frame S' ,

$$\text{Proper Time, } t_0 = t_2' - t_1' \quad \text{--- (1)}$$

In frame S ,

$$\text{Moving Time, } t = t_2 - t_1 \quad \text{--- (2)}$$

$$\Rightarrow t = \frac{t_2' + vx'/c^2 - t_1' - vx'/c^2}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow t = \frac{t_2' - t_1'}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow t = \frac{t_0}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow [t > t_0]$$

∴ The moving clock appears to be slowed down to an stationary observer.

At low velocity, $\frac{v^2}{c^2} \rightarrow 0$

$$[t = t_0]$$

Example: Meson Decay

Meson particles are created when cosmic rays enter the atmosphere. They have velocity 2.994×10^8 m/s and life-time 2 ms (2×10^{-6} s). After that they decay in electrons.

The maximum distance travel by them is,
distance = $2.994 \times 10^8 \times 2 \times 10^{-6} \approx 600$ m

But they are present at sea level due to time dilation effect.

$$t = \frac{t_0}{\sqrt{1 - v^2/c^2}} = \frac{2 \times 10^{-6}}{\sqrt{1 - (2.994 \times 10^8 / 3 \times 10^8)^2}} = 31.7 \times 10^{-6} \text{ sec}$$

distance = $2.994 \times 10^8 \times 31.7 \times 10^{-6} \approx 9,500$ m.

Q. Show from Lorentz Transformation that two events simultaneous ($t_1 = t_2$) at different position ($x_1 \neq x_2$) in a frame S are not, in general, simultaneous in another frame S' ?
[Concept of Simultaneity]

Solⁿ:

$$t_1 = t_2$$

$$x_1 \neq x_2$$

To show, $t'_1 \neq t'_2$

The Lorentz Transformation Equation is,

$$t'_1 = \frac{t_1 - vx_1/c^2}{\sqrt{1 - v^2/c^2}}$$

$$t'_2 = \frac{t_2 - vx_2/c^2}{\sqrt{1 - v^2/c^2}}$$

Now,

$$t'_1 - t'_2 = \frac{-v/c^2 (x_1 - x_2)}{\sqrt{1 - v^2/c^2}} \quad [\because t_1 = t_2]$$

$$\because x_1 \neq x_2 \Rightarrow t'_1 - t'_2 \neq 0$$

$$\therefore t'_1 \neq t'_2$$

Q. At what speed should a clock be moved so that it may appear to loose one minute in each hour?

Solⁿ:

$$t = 60 \text{ sec}$$

$$t_0 = 59 \text{ sec}$$

We have,

$$t = \frac{t_0}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow 1 - \frac{v^2}{c^2} = \left(\frac{t_0}{t}\right)^2 = \left(\frac{59}{60}\right)^2 = 0.96694$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - 0.96694 = 0.03305$$

$$\Rightarrow v = 3 \times 10^8 \times \sqrt{0.03305} = 5.4 \times 10^7 \text{ m/s.}$$

Q. At what speed should a clock be moved so that it may appear to loose four minute in 24 hour?

Solⁿ:

$$t = 24 \times 60 = 1440 \text{ min}$$

$$t_0 = 1440 - 4 = 1436 \text{ min}$$

$$\frac{v^2}{c^2} = 1 - \left(\frac{t_0}{t}\right)^2 = 1 - \left(\frac{1436}{1440}\right)^2 = 1 - 0.99445 = 5.5 \times 10^{-3}$$

$$\Rightarrow v = 3 \times 10^8 \times \sqrt{5.5 \times 10^{-3}} = 2.22 \times 10^7 \text{ m/sec.}$$

Velocity Addition Theorem

Consider two inertial frame S & S' . A particle 'P' has velocity u' w.r.to frame S' and velocity u w.r.to frame S . The three components of velocity are given as:-

In frame S ,

$$u_x = \frac{dx}{dt}, \quad u_y = \frac{dy}{dt} \quad \& \quad u_z = \frac{dz}{dt}$$

In frame S' ,

$$u'_x = \frac{dx'}{dt'}, \quad u'_y = \frac{dy'}{dt'} \quad \& \quad u'_z = \frac{dz'}{dt'}$$

Inverse Lorentz Equation in differential form,

$$dx = \frac{dx' + v dt'}{\sqrt{1 - v^2/c^2}}, \quad dt = \frac{dt' + \frac{v}{c^2} dx'}{\sqrt{1 - v^2/c^2}}$$

$$dy = dy'$$

$$dz = dz'$$

We have,

$$u_x = \frac{dx}{dt} = \frac{dx' + v dt'}{dt' + \frac{v}{c^2} dx'} = \frac{dt' (dx'/dt' + v)}{dt' (1 + \frac{v}{c^2} \frac{dx'}{dt'})}$$

$$\Rightarrow \boxed{u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} u'_x}}$$

$$u_y = \frac{dy}{dt} = \frac{dy' \sqrt{1 - v^2/c^2}}{dt' + v/c^2 dx'}$$

$$\Rightarrow u_y = \frac{dy' \sqrt{1 - v^2/c^2}}{dt' (1 + \frac{v}{c^2} dx'/dt')}$$

$$\Rightarrow \boxed{u_y = \frac{u'_y \sqrt{1 - v^2/c^2}}{1 + \frac{v}{c^2} u'_x}}$$

Also,

$$u_z = \frac{dz}{dt} = \frac{dz' \sqrt{1 - v^2/c^2}}{dt' + v/c^2 dx'}$$

$$\Rightarrow u_z = \frac{dz' \sqrt{1 - v^2/c^2}}{dt' (1 + \frac{v}{c^2} dx'/dt')}$$

$$\Rightarrow \boxed{u_z = \frac{u'_z \sqrt{1 - v^2/c^2}}{1 + \frac{v}{c^2} u'_x}}$$

Q. Show that Velocity Addition Theorem is consistent with Einstein's Second Postulate?

We have,

$$u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} u'_x}$$

Let, $u'_x = c$

$$u_x = \frac{c + v}{1 + \frac{v}{c^2} \cdot c} = \frac{c + v}{(c + v)/c} = c.$$

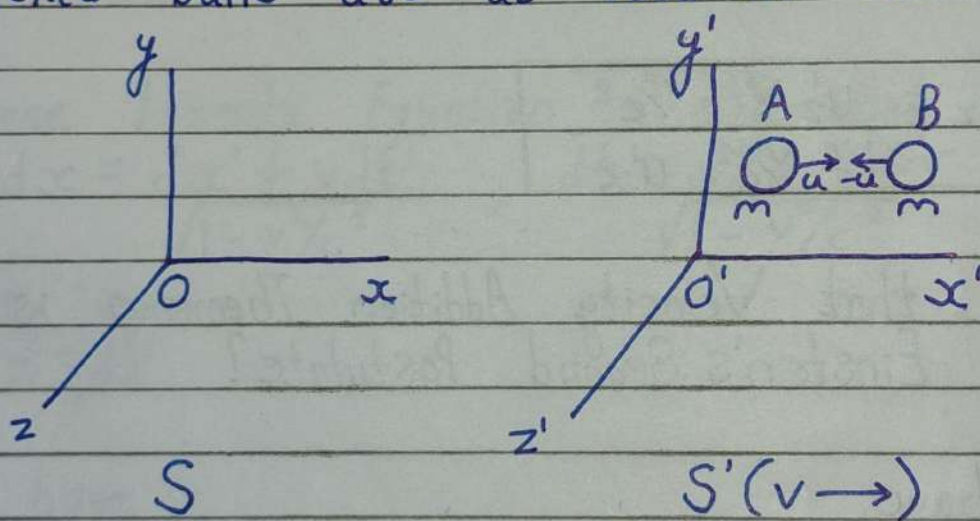
Imp. # Variation of Mass with Velocity

Consider two inertial frame S & S' . S is at rest and S' is moving with constant velocity ' v ' in the x -direction. Two exactly similar balls A & B having equal mass m , approach each other at equal speeds (in frame S'). They collide with each other and stick into one body.

Considering Conservation of Momentum,

$$m \cdot u + m(-u) = (m+m) \times \text{velocity} \\ \Rightarrow \text{velocity} = 0$$

∴ Sticked balls are at rest in frame S' .



Now, consider the collision w.r.to frame S . Mass and velocity of A is m_1 & u_1 . Similarly, mass and velocity of ball B is m_2 & u_2 before collision. After collision, mass becomes $(m_1 + m_2)$ and velocity ' v '.

Applying Conservation of Momentum,

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v \quad \text{--- (1)}$$

Using velocity addition theorem, $u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} \cdot u'_x}$

$$\text{For ball A, } u_1 = \frac{u + v}{1 + \frac{vu}{c^2}}$$

$$\text{For ball B, } u_2 = \frac{-u + v}{1 - \frac{vu}{c^2}}$$

Putting these values in eqⁿ. (1),

$$m_1 \left(\frac{u + v}{1 + \frac{vu}{c^2}} \right) + m_2 \left(\frac{v - u}{1 - \frac{vu}{c^2}} \right) = m_1 v + m_2 v$$

$$\Rightarrow m_1 \left[\frac{u + v}{1 + \frac{uv}{c^2}} - v \right] = m_2 \left[v - \frac{v - u}{1 - \frac{uv}{c^2}} \right]$$

$$\Rightarrow \left[\frac{\cancel{u} + \cancel{v} - \cancel{v} - \frac{uv^2}{c^2}}{1 + \frac{uv}{c^2}} \right] m_1 = \left[\frac{\cancel{v} - \frac{uv^2}{c^2} - \cancel{v} + \cancel{u}}{1 - \frac{uv}{c^2}} \right] m_2$$

$$\Rightarrow \frac{m_1}{m_2} = \frac{1 + \frac{uv}{c^2}}{1 - \frac{uv}{c^2}} \quad \text{--- (2)}$$

Let us consider,

$$1 - \frac{u_1^2}{c^2} = 1 - \frac{(u+v)^2}{c^2(1+uv/c^2)^2}$$

$$= 1 - \frac{(u/c + v/c)^2}{(1+uv/c^2)^2}$$

$$= \frac{(1+uv/c^2)^2 - (u/c + v/c)^2}{(1+uv/c^2)^2}$$

$$= \frac{1 + u^2v^2/c^4 + \cancel{2uv/c^2} - u^2/c^2 - v^2/c^2 - \cancel{2uv/c^2}}{(1+uv/c^2)^2}$$

$$= \frac{(1 - u^2/c^2) - v^2/c^2(1 - u^2/c^2)}{(1+uv/c^2)^2}$$

$$\Rightarrow 1 - \frac{u_1^2}{c^2} = \frac{(1 - u^2/c^2)(1 - v^2/c^2)}{(1+uv/c^2)^2} \quad \text{———— (3)}$$

Similarly,

$$1 - \frac{u_2^2}{c^2} = \frac{(1 - u^2/c^2)(1 - v^2/c^2)}{(1-uv/c^2)^2} \quad \text{———— (4)}$$

Dividing eqⁿ. (4) by eqⁿ. (3),

$$\frac{1 - u_2^2/c^2}{1 - u_1^2/c^2} = \left(\frac{1+uv/c^2}{1-uv/c^2} \right)^2 \quad \text{———— (5)}$$

Using eqⁿ. (2) & eqⁿ. (5),

$$\frac{1 - u_2^2/c^2}{1 - u_1^2/c^2} = \frac{m_1^2}{m_2^2}$$

$$\Rightarrow m_2^2 \left(1 - \frac{u_2^2}{c^2} \right) = m_1^2 \left(1 - \frac{u_1^2}{c^2} \right)$$

$$\Rightarrow m_2 \sqrt{1 - \frac{u_2^2}{c^2}} = m_1 \sqrt{1 - \frac{u_1^2}{c^2}} \quad \text{———— (6)}$$

Since, m_1 , m_2 , u_1 & u_2 are independent of each other. Therefore, above condition is true only if both LHS & RHS are equal to a constant m_0 (Rest Mass).

$$\text{i.e. } m_1 \sqrt{1 - \frac{u_1^2}{c^2}} = m_0$$

$$\& \quad m_2 \sqrt{1 - \frac{u_2^2}{c^2}} = m_0$$

In general, for a body of mass ' m ' and velocity ' v ,'

$$m \sqrt{1 - \frac{v^2}{c^2}} = m_0$$

$$\boxed{m = \frac{m_0}{\sqrt{1 - v^2/c^2}}} \quad \text{———— (7).}$$

Case-1: $v \ll c$

$$\frac{v^2}{c^2} \rightarrow 0 \Rightarrow m = m_0.$$

Case-2: $v = c$

$$m = \frac{m_0}{\sqrt{1 - c^2/c^2}} = \infty$$

$\Rightarrow$ No material particle can move with speed of light.

V. Imp # Mass-Energy Equivalence

Consider a particle at rest having rest-mass m_0 . A force F is applied on it. It moves distance dx with velocity v .

According to Work-Energy Theorem,

$$\begin{aligned} \text{Change in K.E.} &= \text{Work done} \\ \Rightarrow dK &= F \cdot dx \quad \text{--- (1)} \end{aligned}$$

From Newton's Second Law,

$$F = \frac{dp}{dt} = \frac{d}{dt}(mv)$$

$$\Rightarrow F = m \frac{dv}{dt} + v \frac{dm}{dt}$$

Putting this value of F in eqⁿ. (1),

$$dK = m \frac{dv}{dt} \cdot dx + v \frac{dm}{dt} \cdot dx$$

$$\Rightarrow dK = mv \cdot dv + v^2 \cdot dm \quad \text{--- (2)} \quad \left[\frac{dx}{dt} = v \right]$$

$$\therefore m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow m^2 = \frac{m_0^2}{1 - v^2/c^2}$$

$$\Rightarrow m^2 - \frac{m^2 v^2}{c^2} = m_0^2$$

$$\Rightarrow m^2 c^2 - m^2 v^2 = m_0^2 c^2$$

$$\Rightarrow m^2 c^2 = m^2 v^2 + m_0^2 c^2$$

In differential form,

$$c^2 \cdot 2m \cdot dm = m^2 \cdot 2v \cdot dv + v^2 \cdot 2m \cdot dm + 0$$

$$\Rightarrow c^2 dm = m v \cdot dv + v^2 dm \quad \text{———— (3)}$$

From eqⁿ. (2) & (3),

$$dK = c^2 dm$$

$$\Rightarrow \int_0^K dK = \int_{m_0}^m c^2 dm$$

$$\Rightarrow [K]_0^K = c^2 [m]_{m_0}^m$$

$$\Rightarrow \boxed{K = (m - m_0) c^2}, \text{ Relativistic K.E.}$$

$$\text{Rest-mass energy} = m_0 c^2 \quad [\text{P.E.}]$$

So, Total Energy = K.E. + Rest Energy

$$\Rightarrow E = (m - m_0) c^2 + m_0 c^2$$

$$\Rightarrow \boxed{E = mc^2}, \text{ Mass-Energy Equivalence.}$$

Experimental Verification:

1. Pair Annihilation:

When an electron and a positron combine with each other, the entire mass is converted into energy and we get two gamma-ray photons.

2. Pair Production:

When a gamma-ray photon of energy 1.02 MeV comes near a nucleus, the energy is converted into mass. A pair of electron and positron is produced.

Relation Between Total Energy & Momentum

$$E = mc^2$$

$$\& E = \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow E^2 \left(1 - \frac{v^2}{c^2}\right) = m_0^2 c^4$$

$$\Rightarrow E^2 - \frac{E^2 v^2}{c^2} = m_0^2 c^4$$

$$\Rightarrow E^2 - \frac{m^2 c^4 v^2}{c^2} = m_0^2 c^4$$

$$\Rightarrow E^2 - p^2 c^2 = m_0^2 c^4 \Rightarrow \boxed{E^2 = p^2 c^2 + m_0^2 c^4}$$

Q. Show that at low velocity, the Relativistic K.E. reduces to the classical formula?

Ans:

We have,

$$K = (m - m_0)c^2$$

$$\Rightarrow K = \left(\frac{m_0}{\sqrt{1 - v^2/c^2}} - m_0 \right) c^2$$

$$\Rightarrow K = \left[\left(1 - \frac{v^2}{c^2} \right)^{-1/2} - 1 \right] m_0 c^2$$

Applying Binomial Theorem and Neglecting higher order terms because $v \ll c$.

$$K = \left[1 + \frac{1}{2} \cdot \frac{v^2}{c^2} - 1 \right] m_0 c^2$$

$$\Rightarrow K = \frac{1}{2} m_0 v^2.$$

Q. Show that the momentum of particle of rest-mass m_0 and Kinetic energy K.E. is given by,

$$p = \sqrt{\frac{KE^2}{c^2} + 2m_0 KE}.$$

Ans:

We have,

$$E^2 = p^2 c^2 + m_0^2 c^4$$

$$\Rightarrow (KE + m_0 c^2)^2 = p^2 c^2 + m_0^2 c^4$$

$$\Rightarrow KE^2 + m_0^2 c^4 + 2KE \cdot m_0 c^2 = p^2 c^2 + m_0^2 c^4$$

$$\Rightarrow p^2 = \frac{KE^2}{c^2} + 2m_0 \cdot KE$$

$$\Rightarrow p = \sqrt{\frac{KE^2}{c^2} + 2m_0 \cdot KE} \quad \text{. proved}$$

Numericals

Q. A man weight 50 Kg on Earth. When he is in rocket ship in flight, his mass is 50.5 Kg as measured by an observer on Earth. What is the speed of rocket?

Solⁿ:

$$m_0 = 50 \text{ Kg}$$

$$m = 50.5 \text{ Kg}$$

$$v = ?$$

We know,

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow m^2 = \frac{m_0^2}{(\sqrt{1 - v^2/c^2})^2}$$

$$\Rightarrow m^2 - \frac{m^2 v^2}{c^2} = m_0^2$$

$$\Rightarrow 1 - \frac{v^2}{c^2} = \frac{m_0^2}{m^2}$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \frac{m_0^2}{m^2}$$

$$\Rightarrow v = c \sqrt{1 - \frac{m_0^2}{m^2}} = 3 \times 10^8 \sqrt{1 - \frac{50^2}{50.5^2}} = 4.21 \times 10^7 \text{ m/s.}$$

Imp. Q. If the total energy of a particle is exactly thrice of its rest energy, what is the velocity of particle?

Solⁿ:

Given,

$$\text{Total Energy} = 3 \times \text{Rest Energy}$$

$$\Rightarrow m_e c^2 = 3 m_0 c^2$$

$$\Rightarrow \frac{m_0}{\sqrt{1 - v^2/c^2}} = 3 m_0$$

$$\Rightarrow \sqrt{1 - \frac{v^2}{c^2}} = \frac{1}{3}$$

$$\Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{9}$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \frac{1}{9} = \frac{8}{9}$$

$$\Rightarrow v = c \sqrt{\frac{8}{9}}$$

$$\Rightarrow v = 3 \times 10^8 \sqrt{\frac{8}{9}} = 2.82 \times 10^8 \text{ m/s.}$$

Imp Q. If a K.E. of a body is twice its rest energy, find the velocity?

Solⁿ:

Given,

$$K.E. = 2 \times \text{Rest Energy}$$

$$\Rightarrow (m - m_0)c^2 = 2m_0c^2$$

$$\Rightarrow m = 3m_0$$

$$\Rightarrow \frac{m_0}{\sqrt{1 - v^2/c^2}} = 3m_0$$

$$\Rightarrow \sqrt{1 - \frac{v^2}{c^2}} = \frac{1}{3}$$

$$\Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{9}$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \frac{1}{9} = \frac{8}{9}$$

$$\Rightarrow v = c \sqrt{\frac{8}{9}}$$

$$\Rightarrow v = 2.82 \times 10^8 \text{ m/s.}$$

9. The mass of a moving electron is 11 times its rest mass, find its K.E. and momentum?

Solⁿ:

$$m = 11 m_0$$

We know,

$$K.E. = (m - m_0)c^2 = (11m_0 - m_0)c^2 = 10m_0c^2$$

$$\Rightarrow K.E. = 10 \times 9.1 \times 10^{-31} \times 9 \times 10^{16}$$

$$\Rightarrow K.E. = 8.19 \times 10^{-13} \text{ J.}$$

Now,

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow 1 - \frac{v^2}{c^2} = \left(\frac{m_0}{m}\right)^2 = \left(\frac{m_0}{11m_0}\right)^2$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \left(\frac{1}{11}\right)^2 = 1 - \frac{1}{121} = \frac{120}{121}$$

$$\Rightarrow v = c \sqrt{\frac{120}{121}} = 2.98 \times 10^8 \text{ m/s}$$

$$\therefore p = mv = 11m_0 v = 11 \times 9.1 \times 10^{-31} \times 2.98 \times 10^8$$

$$\Rightarrow p = 2.98 \times 10^{-21} \text{ kg m/s.}$$

Q. Find the mass and speed of 2 MeV electron?

Solⁿ:

$$E = 2 \text{ MeV} = 2 \times 10^6 \times 1.6 \times 10^{-19} = 3.2 \times 10^{-13} \text{ J}$$

$$m = ?$$

$$v = ?$$

We know,

$$E = mc^2$$

$$\Rightarrow m = \frac{E}{c^2} \quad \# \quad \cancel{2.9 \times 10^{-30}}$$

$$\Rightarrow m = 3.55 \times 10^{-30} \text{ kg}$$

Now,

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow 1 - \frac{v^2}{c^2} = \left(\frac{m_0}{m} \right)^2$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \left(\frac{m_0}{m} \right)^2$$

$$\Rightarrow v = c \sqrt{1 - \left(\frac{m_0}{m} \right)^2} = 3 \times 10^8 \sqrt{1 - \left(\frac{9.1 \times 10^{-31}}{3.55 \times 10^{-30}} \right)^2}$$

$$\Rightarrow v = 2.89 \times 10^8 \text{ m/s.}$$

Q. A particle of rest mass m_0 moves with speed $\frac{c}{\sqrt{2}}$. Calculate its mass, momentum, total energy and K.E.?

Solⁿ:

$$v = \frac{c}{\sqrt{2}}$$

Now,

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}} = \frac{m_0}{\sqrt{1 - (\cancel{c}/\sqrt{2}\cancel{c})^2}} = 1.414 m_0 = \sqrt{2} m_0.$$

$$p = mv = \sqrt{2} m_0 \cdot \frac{c}{\sqrt{2}} = m_0 c.$$

$$E = mc^2 = \sqrt{2} m_0 c^2.$$

$$K.E. = (m - m_0)c^2 = (\sqrt{2} m_0 - m_0)c^2 = 0.414 m_0 c^2.$$

Q. How much does a proton gain in mass when accelerated to a K.E. of 500 MeV?

Solⁿ:

$$K.E. = 500 \text{ MeV}$$

$$\Rightarrow K.E. = 500 \times 10^6 \times 1.6 \times 10^{-19} = 8 \times 10^{-11} \text{ J}$$

$$\Rightarrow (m - m_0) c^2 = 8 \times 10^{-11}$$

$$\Rightarrow m - m_0 = \frac{8 \times 10^{-11}}{9 \times 10^{16}} = 8.88 \times 10^{-28} \text{ Kg.}$$

Q. How fast must an electron move in order that its mass equals the rest mass of proton?

Solⁿ:

$$m_0 = 9.1 \times 10^{-31} \text{ Kg}$$

$$m = 1.67 \times 10^{-27} \text{ Kg}$$

$$v = ?$$

We know,

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow 1 - \frac{v^2}{c^2} = \left(\frac{m_0}{m} \right)^2$$

$$\Rightarrow v = c \sqrt{1 - \left(\frac{m_0}{m} \right)^2} = 2.99 \times 10^8 \text{ m/s.}$$

Q. Calculate the amount of work to be done to increase the speed of an electron from $0.6c$ to $0.8c$? Given that, the rest energy of electron is 0.5 MeV .

Solⁿ:

$$v_1 = 0.6c$$

$$v_2 = 0.8c$$

$$m_0 c^2 = 0.5 \text{ MeV} = 0.5 \times 10^6 \times 1.6 \times 10^{-19} = 8 \times 10^{-14} \text{ J}$$

Now,

$$K_1 = (m_1 - m_0) c^2$$

$$\Rightarrow K_1 = \left(\frac{m_0}{\sqrt{1 - v_1^2/c^2}} - m_0 \right) c^2$$

$$\Rightarrow K_1 = \left(\frac{1}{\sqrt{1 - \left(\frac{0.6c}{c}\right)^2}} - 1 \right) m_0 c^2$$

$$\Rightarrow K_1 = 0.25 m_0 c^2$$

Again,

$$K_2 = (m_2 - m_0) c^2$$

$$\Rightarrow K_2 = \left(\frac{m_0}{\sqrt{1 - v_2^2/c^2}} - m_0 \right) c^2$$

$$\Rightarrow K_2 = \left(\frac{1}{\sqrt{1 - \left(\frac{0.8c}{c}\right)^2}} - 1 \right) m_0 c^2$$

$$\Rightarrow K_2 = 0.66 m_0 c^2$$

$\therefore$ Work done = Change in K.E.

$$\Rightarrow \text{Work done} = K_2 - K_1 = 0.416 m_0 c^2$$

$$\Rightarrow \text{Work done} = 0.416 \times 8 \times 10^{-14}$$

$$\Rightarrow \text{Work done} = 3.328 \times 10^{-14} \text{ J.}$$

Q. Verify the relation $\frac{1}{\sqrt{1 - v^2/c^2}} = 1 + \frac{\text{K.E.}}{m_0 c^2}$.

Solⁿ:

We have,

$$\text{K.E.} = (m - m_0) c^2$$

$$\Rightarrow \text{K.E.} = \left(\frac{m_0}{\sqrt{1 - v^2/c^2}} - m_0 \right) c^2$$

$$\Rightarrow \text{K.E.} = \left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right) c^2 m_0$$

$$\Rightarrow \frac{\text{K.E.}}{m_0 c^2} = \frac{1}{\sqrt{1 - v^2/c^2}} - 1$$

$$\Rightarrow \frac{1}{\sqrt{1 - v^2/c^2}} = 1 + \frac{\text{K.E.}}{m_0 c^2} \quad \text{proved}$$

Q. Show that the relativistic form of Newton's Second Law, when force F is parallel to velocity v , is $F = m_0 \frac{d\vec{v}}{dt} \left(1 - \frac{v^2}{c^2}\right)^{-3/2}$.

Solⁿ:

We know,

$$F = \frac{dp}{dt} = \frac{d}{dt}(mv)$$

$$= \frac{d}{dt} \left(\frac{m_0 v}{\sqrt{1 - v^2/c^2}} \right)$$

$$= m_0 \cdot \frac{d}{dt} \left[v \cdot \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \right]$$

$$= m_0 \left[\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \cdot \frac{d\vec{v}}{dt} + v \cdot \frac{1}{2} \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \cdot \frac{2v}{c^2} \frac{d\vec{v}}{dt} \right]$$

$$= m_0 \frac{d\vec{v}}{dt} \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \left[\left(1 - \frac{v^2}{c^2}\right) + \frac{v^2}{c^2} \right]$$

$$\Rightarrow F = m_0 \frac{d\vec{v}}{dt} \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \quad \underline{\text{proved}}$$

Q. Show that Space-Time interval is invariant under Lorentz Transformation?

Solⁿ:

The space-time interval is,
 $x^2 + y^2 + z^2 - c^2 t^2$

let,
 $x'^2 + y'^2 + z'^2 - c^2 t'^2$

$$= \frac{x - vt}{\sqrt{1 - v^2/c^2}} + y^2 + z^2 - \frac{c^2 (t - vx/c^2)^2}{1 - v^2/c^2}$$

$$= \frac{(x - vt)^2}{1 - v^2/c^2} + y^2 + z^2 - \frac{c^2 (t - vx/c^2)^2}{1 - v^2/c^2}$$

$$= \frac{(x - vt)^2 - c^2 (t - vx/c^2)^2}{1 - v^2/c^2} + y^2 + z^2$$

$$= \frac{x^2 - 2xvt + v^2 t^2 - c^2 t^2 + 2vxt - v^2 x^2/c^2}{1 - v^2/c^2} + y^2 + z^2$$

$$= \frac{x^2(1 - v^2/c^2) - c^2 t^2(1 - v^2/c^2)}{1 - v^2/c^2} + y^2 + z^2$$

$$= x^2 - c^2 t^2 + y^2 + z^2 \text{ proved}$$